

Numerical Solutions to Problems in Electrochemistry



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A dissertation submitted for the degree of
Masters in Mathematical Sciences

Trinity 2026

Abstract

This is a project on PINNs that do not seem to work very well.

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Chapter 1

Introduction

Electrochemistry is a branch of chemistry that is concerned with the relationship between electrical applied potential and chemical change. These reactions arise at the interface between an electronic conductor and an ionic conductor, where charge transfer drives oxidation and reduction of chemical species. The study of these reactions has applications in biochemical sensing, energy storage, technology and many others that underpin much of modern biology and nanotechnology. In particular, electroanalytical techniques have enabled the detection of trace analytes in biological fluids, the development of rechargeable batteries and fuel cells, and the fabrication of nanoscale electronic devices, underlining the broad relevance of electrochemical science across disciplines.

Voltammetry is a common technique to analyse an electroactive species in solution. Voltammetry measures the current that is induced into an electrochemical circuit as the potential of the working electrode is varied with respect to the reference electrode. The current response arises from Faradaic processes, in which electrons are transferred between the electrode surface and the electroactive species, and its magnitude reflects both the kinetics of this charge transfer and the rate at which species are transported to the electrode. Voltammetric techniques are distinguished by how the applied potential is varied over time. In cyclic voltammetry, the potential is swept linearly between two limits and then reversed, producing a characteristic current–potential curve from which thermodynamic and kinetic parameters may be inferred. Other waveforms, such as the sinusoidal perturbations of electrochemical impedance spectroscopy or the staircase profiles of square-wave voltammetry, offer complementary sensitivity and resolution.

Voltammetry has been extensively studied mathematically and numerically. The electron transfer kinetics at the electrode surface are described by the Butler–Volmer

model, which relates the rate of the electrode reaction to the applied overpotential through exponential anodic and cathodic contributions, or alternatively by the Marcus–Hush framework, which grounds the kinetics in a quantum-mechanical treatment of solvent reorganisation. At equilibrium, the Nernst equation dictates the ratio of oxidised to reduced species concentrations at the electrode surface as a function of the applied potential. The transport of electroactive species through the solution is governed by Fick’s laws of diffusion, which, combined with the electrode boundary conditions described above, yield a partial differential equation (PDE) that fully characterises the evolution of the concentration profile in the cell. The solution of this system—the so-called forward problem—predicts the current response given a known set of physical parameters, and has been extensively studied both analytically and numerically [2].

We are interested in the inverse of this process: given the measured current response from a voltammetric experiment, we wish to infer the underlying physical parameters that characterise the electroactive species in solution. These parameters may include the formal potential, diffusion coefficients, and electron transfer rate constants, knowledge of which provides quantitative insight into the thermodynamics and kinetics of the redox system under study. This constitutes the inverse problem, and its solution is considerably more challenging than the forward problem. The mapping from parameters to current is nonlinear and the measurements are corrupted by noise, meaning that naive inversion is typically ill-posed and that uncertainty in the inferred parameters must be carefully quantified.

Many different techniques have been employed to solve this inverse problem. Deterministic optimisation methods, such as least-squares fitting, provide point estimates of the parameters but offer no principled account of uncertainty. Bayesian inference addresses this limitation by treating the parameters as random variables and seeking their posterior distribution conditioned on the observed data. Markov chain Monte Carlo (MCMC) methods have emerged as the dominant computational approach to posterior sampling in electrochemistry, enabling rigorous uncertainty quantification without requiring analytical tractability of the posterior [5]. However, standard MCMC algorithms such as the random-walk Metropolis–Hastings sampler can mix poorly in high-dimensional or geometrically complex parameter spaces, and the requirement to repeatedly solve the forward PDE makes the burn-in phase computationally expensive.

With the advent of modern automatic differentiation (AD) software, it is now possible to efficiently compute exact derivatives of arbitrary numerical programs,

including complete PDE solvers, with respect to their inputs [1]. This capability opens new avenues for gradient-based inference in electrochemistry. In particular, the gradient of the simulated current with respect to the model parameters can be obtained at a cost comparable to a single forward solve, enabling the use of gradient-informed algorithms both during the burn-in phase and throughout the sampling procedure. During burn-in, gradient information can be exploited to drive the chain rapidly toward regions of high posterior probability, dramatically reducing the number of forward solves required before convergence. Once in the posterior, gradient-based samplers such as Hamiltonian Monte Carlo (HMC) use the gradient to make more informed proposals, resulting in higher acceptance rates and faster exploration of the posterior landscape compared with gradient-free alternatives.

This dissertation demonstrates the applicability of gradient-based methodologies to solve the inverse problem in electrochemical experiments. We develop a differentiable PDE solver for common voltammetric waveforms, validate it against known analytical results, and integrate it with gradient-informed MCMC algorithms to perform Bayesian inference of electrochemical parameters. We show that the use of automatic differentiation yields substantial improvements in computational efficiency over gradient-free methods, and we illustrate the approach on both synthetic and experimental data.

Chapter 2

Electrochemical Systems

A typical electrochemical experiment uses an electrochemical cell to drive reduction and oxidation (redox) reactions at the surface of a working electrode. The cell consists of at least two electrodes immersed in an electrolyte solution: the *working electrode*, at whose surface the reaction of interest takes place, and a *counter electrode*, which completes the circuit. A third *reference electrode* is commonly included to provide a stable, known potential against which the potential of the working electrode is measured and controlled. A device called a *potentiostat* maintains the desired potential difference between the working and reference electrodes whilst measuring the current that flows between the working and counter electrodes.

A one-electron redox reaction can be written as:



where species A is reduced – that is, it gains one electron – to form species B. The reverse process, in which B loses an electron to regenerate A, is the corresponding oxidation.

2.1 Voltammetry

Voltammetry is the class of electroanalytical techniques in which the potential applied to the working electrode is varied in a prescribed manner and the resulting current is recorded. Because the current is directly related to the rate of the electrochemical reaction at the electrode surface, the current–potential curve – called a *voltammogram* – encodes information about the thermodynamics and kinetics of the redox process as well as the transport of chemical species in solution.

In a cyclic voltammetry (CV) experiment the applied potential is swept linearly with time from some starting potential E_i , at which the species A is stable and no

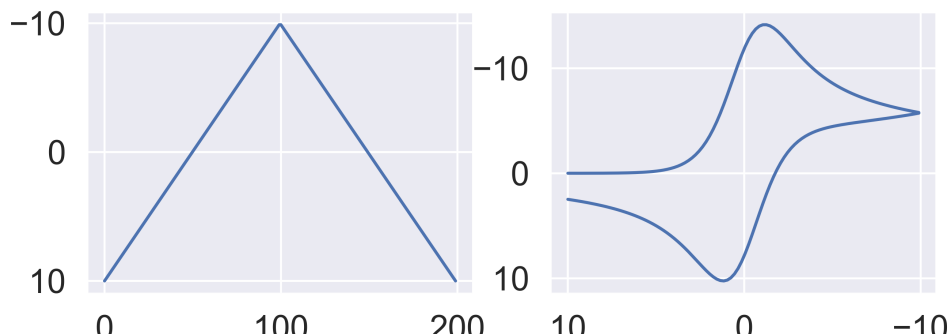


Figure 2.1: Left: the triangular potential waveform applied in a cyclic voltammetry experiment. Right: the resulting voltammogram showing cathodic (reduction) and anodic (oxidation) current peaks.

significant reaction occurs, to a vertex potential E_v , at which electron transfer occurs rapidly to form species B. The potential is then swept back to E_i , causing electron transfer in the opposite direction and the reformation of A. This triangular waveform is illustrated in the left panel of Figure 2.1.

Throughout this process the current I is recorded. Plotting the current against the applied potential gives the voltammogram shown in the right panel of Figure 2.1. The forward sweep produces a reduction (cathodic) peak as the potential moves through the range at which A is converted to B; the return sweep produces an oxidation (anodic) peak as B is reconverted to A. The peak shape arises from competition between the electrode kinetics, which accelerate as the driving potential increases, and the depletion of A near the electrode surface caused by the ongoing reaction.

This technique is particularly useful because the position, shape, and relative heights of the peaks provide a signature of the electrochemical process. The potential at which each peak occurs is related to the standard reduction potential E° , and the separation between the cathodic and anodic peaks carries information about the electrode kinetics, as discussed in section 2.2.

The *scan rate* ν (V s^{-1}) is the constant rate at which the potential is swept, defined as

$$\nu = \frac{dE}{dt}. \quad (2.2)$$

The scan rate is an important experimental parameter: faster scans probe shorter timescales and can resolve kinetic processes that would otherwise reach equilibrium on a slower scan, whilst slower scans allow more time for diffusion and can reveal coupled chemical reactions.

2.2 Electrode Kinetics

In order to model the reaction (2.1) in a voltammetry experiment we require a description of how the applied potential drives the interconversion of A and B at the electrode surface. The electrode reaction rate is commonly expressed through the *flux* J ($\text{mol m}^{-2} \text{s}^{-1}$) of material consumed or produced at the surface.

2.2.1 The Nernst Equation: Reversible Kinetics

When the rate of electron transfer is very fast relative to the timescale of the experiment, the concentrations of A and B at the electrode surface are in thermodynamic equilibrium with the applied potential at every instant. In this *reversible* (or *Nernstian*) limit, the surface concentrations $c_A(0, t)$ and $c_B(0, t)$ are related to the applied potential $E(t)$ by the Nernst equation,

$$E(t) = E^\circ + \frac{RT}{F} \ln \frac{c_A(0, t)}{c_B(0, t)}, \quad (2.3)$$

where R is the universal gas constant, T is the absolute temperature, and F is Faraday's constant. Equation (2.3) serves as a boundary condition at the electrode surface; combined with the diffusion equation introduced in section 2.3, it fully determines the concentration fields and hence the current in the reversible limit.

2.2.2 The Butler–Volmer Equation: Finite Electrode Kinetics

When the electron-transfer rate is finite, the surface concentrations depart from the Nernstian equilibrium values, and the reaction rate must be modelled explicitly. The standard model for electrode kinetics is the *Butler–Volmer* equation [4], which gives the flux at the electrode surface as

$$J = k_0 \left[c_A(0, t) \exp\left(-\frac{\alpha F}{RT} \eta(t)\right) - c_B(0, t) \exp\left(\frac{(1 - \alpha) F}{RT} \eta(t)\right) \right], \quad (2.4)$$

where k_0 (ms^{-1}) is the standard rate constant, $\alpha \in (0, 1)$ is the transfer coefficient (often taken to be 0.5 for a symmetric energy barrier), and

$$\eta(t) = E(t) - E^\circ \quad (2.5)$$

is the *overpotential*, measuring the deviation of the applied potential from the equilibrium value. The two exponential terms in (2.4) represent the reduction and oxidation

contributions respectively: a negative overpotential ($E < E^\circ$) drives reduction, whilst a positive overpotential drives oxidation.

The Butler–Volmer equation interpolates naturally between two limits. When k_0 is large, the exponential terms force the surface concentrations towards the Nernstian equilibrium values (2.3), recovering reversible behaviour. When k_0 is small, the reaction is kinetically limited and the flux is directly proportional to k_0 , giving *irreversible* behaviour. The current measured in the external circuit is related to the flux by

$$I(t) = FAJ(t), \quad (2.6)$$

where A (m^2) is the electrode area.

2.3 Diffusion

The electrode reaction consumes A and produces B at the surface. Fresh A must therefore be transported from the bulk of the solution to the electrode, and B must be transported away. In quiescent (unstirred) solutions, the dominant transport mechanism is *diffusion*: the net movement of molecules down concentration gradients driven by random thermal motion.

We model each species by a concentration field $c(x, t)$, where x denotes the distance from the electrode surface and t denotes time. For a planar electrode, the evolution of the concentration is governed by *Fick's second law* [3],

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}, \quad (2.7)$$

where D ($\text{m}^2 \text{s}^{-1}$) is the diffusion coefficient of the species. Equation (2.7) is a linear parabolic partial differential equation.

For the couple A/B we therefore have a system of two diffusion equations,

$$\frac{\partial c_A}{\partial t} = D_A \frac{\partial^2 c_A}{\partial x^2}, \quad (2.8)$$

$$\frac{\partial c_B}{\partial t} = D_B \frac{\partial^2 c_B}{\partial x^2}, \quad (2.9)$$

where D_A and D_B are the respective diffusion coefficients. These equations hold on the semi-infinite domain $x > 0$, representing the bulk solution, and are supplemented by initial conditions and boundary conditions. Initially the solution is homogeneous:

$$c_A(x, 0) = c_A^*, \quad c_B(x, 0) = 0, \quad (2.10)$$

where c_A^* is the bulk concentration of A. Far from the electrode the concentrations recover their bulk values,

$$c_A(x, t) \rightarrow c_A^*, \quad c_B(x, t) \rightarrow 0 \quad \text{as } x \rightarrow \infty, \quad (2.11)$$

whilst at the electrode surface ($x = 0$) the diffusive fluxes are coupled to the electrode reaction through flux-conservation conditions. Conservation of matter at the surface requires

$$D_A \frac{\partial c_A}{\partial x} \bigg|_{x=0} = -D_B \frac{\partial c_B}{\partial x} \bigg|_{x=0} = J(t), \quad (2.12)$$

where $J(t)$ is the reaction flux given by the Butler–Volmer equation (2.4) (or, in the reversible limit, determined implicitly by the Nernst equation (2.3) together with the constraint (2.12)).

The complete mathematical description – comprising the two diffusion equations (2.8) – (2.9), the initial and far-field conditions, and the surface boundary conditions – constitutes a well-posed initial-boundary value problem whose solution gives the concentration fields and, via (2.12), the current $I(t)$ as a function of time during a cyclic voltammetry experiment. The remainder of this thesis is concerned with the analysis and numerical solution of this problem.

2.3.1 Dimensionless Coordinates

There are many non-dimensionalisations that can be performed on the described problem. We follow the transformations similar to those done in [2]

Parameter	Normalisation
concentration	$C_j = c_j / c_A^*$
diffusion coefficient	$d_j = D_j / D_A$
spatial coordinate	$X = x / \epsilon$
time	$T = D_A t / \epsilon^2$
potential	$\theta = (F / \mathcal{RT}) (E - E_f^0)$
scan rate	$\sigma = (\epsilon^2 / D_A) (F / \mathcal{RT}) \nu$
current	$J = I / (\pi \epsilon F D_A c_A^*)$

Chapter 3

Numerical Methods

3.1 Finite Difference Methods

3.2 Automatic Differentiation

Chapter 4

Markov chain Monte Carlo and the Inverse Problem

4.1 Metropolis-Hastings

4.2 Random Walk Metropolis-Hastings

4.3 Hamiltonian Monte Carlo

Chapter 5

Electron Transfer Reaction

Chapter 6

Heterogeneous Reaction

Chapter 7

Adsorption Reaction

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